src/dynamics.py · mass_matrix MMA3001 DeepONet project

# mass_matrix

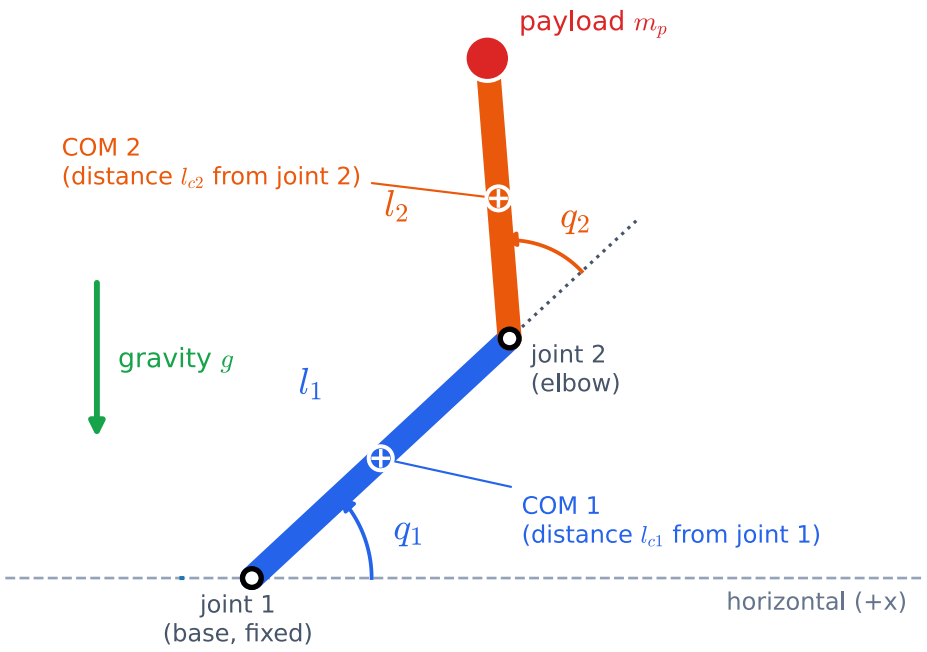
The 2×2 inertia matrix M(q): how torque turns into acceleration, derived from kinetic energy

**In this document**


1. The question
2. Kinematics: positions
3. Velocities
4. Kinetic energy
5. Reading off the matrix
6. Physical meaning of each entry
7. Properties (symmetric, positive definite)
8. Worked example
9. Code-maths dictionary


## The set-up: a two-link robot arm in a vertical plane

Everything in this project describes one physical object: a planar robot arm with two rigid links, two motorised joints, bolted to a wall at the origin. It moves in a *vertical* plane, so gravity matters. At the tip of link 2 we may clip on an extra weight (the **payload**). Each joint has a little friction that slows it down in proportion to how fast it spins.




**The arm.** Blue = link 1, orange = link 2. The angle $q_1$ is measured from the horizontal; the angle $q_2$ is measured from the *straight-on extension* of link 1 (dotted), not from the horizontal. The dots with a white cross are the links' centres of mass (COM).

**Conventions** (fixed everywhere in the project): the state is $x = [q_1, q_2, \dot{q}_1, \dot{q}_2]$; gravity points in the $-y$ direction; potential energy is zero at the height of joint 1; positive angles and torques are anticlockwise.

| Symbol | Meaning | Value in the code | Unit |
|---|---|---|---|
| $l_1,\ l_2$ | link lengths (joint to joint) | 0.5, 0.4 | m |

| | | | |
|---|---|---|---|
| $m_1,\ m_2$ | link masses (without payload) | 2.0, 1.5 | kg |
| $l_{c1},\ l_{c2}$ | distance from a link's joint to its own centre of mass (COM) | 0.25, 0.2 (middle of each link) | m |
| $I_1,\ I_2$ | moment of inertia of a link about its own COM | $ml^2/12$ (uniform rod) | kg m$^2$ |
| $g$ | gravitational acceleration | 9.81 | m/s$^2$ |
| $m_p$ | point-mass payload at the tip of link 2 (varies per sample) | 0 to 1 (dataset), up to 1.5 out-of-range | kg |
| $b$ | viscous joint friction coefficient (varies per sample) | 0.05 to 0.5 | N m s/rad |
| $q_1,\ q_2$ | joint angles: $q_1$ from the horizontal, $q_2$ relative to link 1 | state | rad |
| $\dot{q}_1,\ \dot{q}_2$ | joint angular velocities (a dot means $d/dt$) | state | rad/s |
| $\tau_1,\ \tau_2$ | motor torques applied at joints 1 and 2 (the inputs) | $A\sin(2\pi f t+\phi)$ | N m |

## Toolbox: Lagrangian mechanics in five minutes

You know Newton's second law for a straight-line motion: $F = ma$. For something that *rotates* about a fixed pin the same law reads

$$\tau = I\,\ddot{\theta},$$

where $\tau$ is torque (a twisting push, N m), $I$ the moment of inertia (resistance to being spun, kg m$^2$) and $\ddot{\theta}$ the angular acceleration. For *one* link that is easy. For *two* links, the second link is moved by the first, which is moved by the base, and the joints push on each other with forces we do not know. Newton's method forces us to compute all those internal forces. **Lagrange's trick** avoids them completely by working with *energies*, which are plain numbers.

### The recipe

1. Choose coordinates that describe the pose with no redundancy: here $q_1, q_2$ (the *generalised coordinates*).
2. Compute the kinetic energy $T$ (as a function of $q, \dot{q}$) and the potential energy $V$ (as a function of $q$).
3. For every coordinate $q_k$ evaluate the **Euler-Lagrange equation**

$$\underbrace{\frac{d}{dt}\left(\frac{\partial T}{\partial \dot{q}_k}\right)}_{\text{(a) rate of change of 'momentum'}} - \underbrace{\frac{\partial T}{\partial q_k}}_{\text{(b) pose-dependence of } T} + \underbrace{\frac{\partial V}{\partial q_k}}_{\text{(c) gravity pulls}} = Q_k$$

   where $Q_k$ collects everything we push with from outside (motor torque $\tau_k$, friction).

Reading it: (a) is "how fast is the generalised momentum $\partial T/\partial \dot{q}_k$ changing" (it is the analogue of $d(mv)/dt$); (c) is the force that gravity exerts along coordinate $q_k$, namely minus the slope of the potential energy (a ball rolls downhill, i.e. the force is $-\partial V/\partial q$, so moving it to the other side of the equation gives $+\partial V/\partial q$); (b) is a correction that appears because the "mass" itself can depend on the pose.

The symbol $\partial$ means a **partial derivative**: differentiate with respect to one variable while pretending all the others are constants. For instance if $T = \tfrac{1}{2}(3+\cos q_2)\,\dot{q}_1^2$ then $\partial T/\partial \dot{q}_1 = (3+\cos q_2)\dot{q}_1$ and $\partial T/\partial q_2 = -\tfrac{1}{2}\sin q_2\,\dot{q}_1^2$.

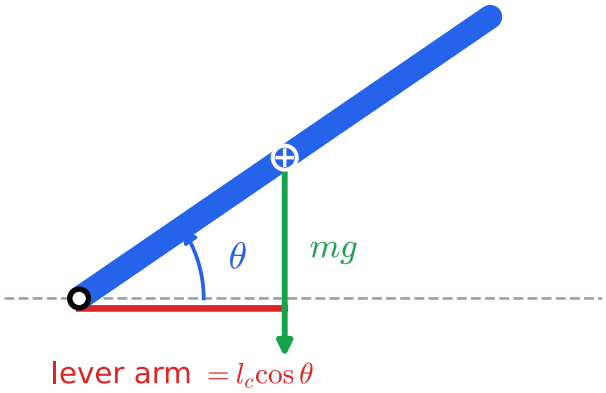
### Test run: one swinging rod (you can check it with Newton)

A uniform rod of mass $m$ pivots at the origin, angle $\theta$ from the horizontal; its COM is a distance $l_c$ from the pivot. Its inertia about the pivot is $I_p = I_{\text{cm}} + ml_c^2$ (explained in the *combine_payload* document). Then

$$T = \tfrac{1}{2} I_p \dot{\theta}^2, \qquad V = mg\, l_c \sin\theta.$$

Euler-Lagrange: $\frac{d}{dt}(I_p\dot{\theta}) - 0 + mgl_c\cos\theta = \tau$, i.e.

$$I_p\ddot{\theta} + mg\, l_c \cos\theta = \tau.$$




**Newton's view of the same rod.** Gravity pulls down with force $mg$ at the COM. Only the horizontal distance $l_c\cos\theta$ to the pivot gives a twist (the lever arm), so gravity's torque about the pin is $mg\, l_c\cos\theta$ - exactly the term Lagrange produced.

Newton gives $I_p\ddot{\theta} = \tau - (\text{gravity torque})$, the same equation. So the method works, and for two links it needs only energies. That is the whole idea; the rest is careful algebra.

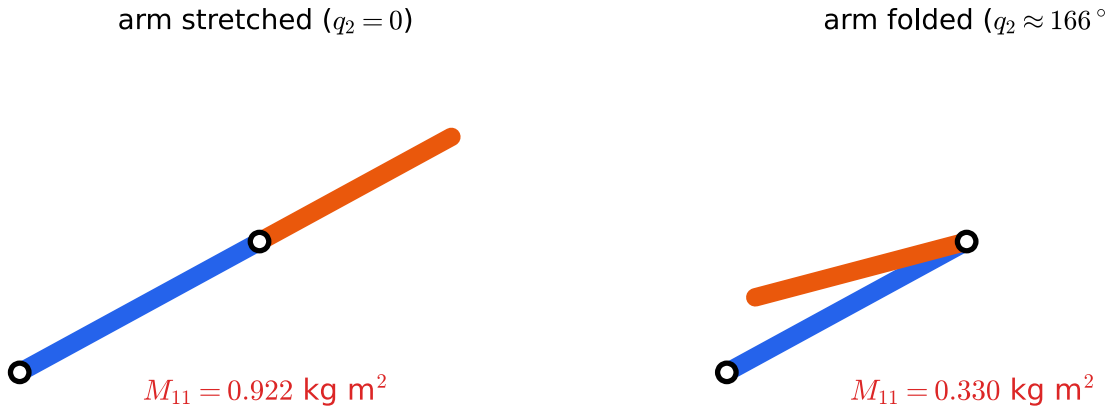
## 1 The question this function answers

If a joint motor applies a torque, how fast does the arm speed up? For a single rotating rod the answer was $\tau = I\ddot{\theta}$: the inertia $I$ is the exchange rate between torque and angular acceleration. For two joints the exchange rate becomes a $2 \times 2$ table, the **mass matrix** (also called inertia matrix) $M(q)$:

$$\begin{pmatrix}\tau_1\\ \tau_2\end{pmatrix} = \underbrace{\begin{pmatrix}M_{11} & M_{12}\\ M_{12} & M_{22}\end{pmatrix}}_{M(q)} \begin{pmatrix}\ddot{q}_1\\ \ddot{q}_2\end{pmatrix} + (\text{velocity terms, gravity, friction}).$$

Two things are new compared with a single rod. First, the table has **off-diagonal entries**: torque at joint 2 can accelerate joint 1 and vice versa (the links are connected!). Second, the entries **depend on the pose** $q$: a stretched arm is much harder to swing than a folded one, exactly like a skater spinning faster when pulling her arms in.

```
def mass_matrix(q, p):
    q1, q2 = q
    m2e, lc2e, I2e = combine_payload(p.m2, p.lc2, p.I2, p.l2, p.m_p)
    c2 = np.cos(q2)
    M11 = p.m1*p.lc1**2 + p.I1 + m2e*(p.l1**2 + lc2e**2 + 2*p.l1*lc2e*c2) + I2e
    M12 = m2e*(lc2e**2 + p.l1*lc2e*c2) + I2e
    M22 = m2e*lc2e**2 + I2e
    return np.array([[M11, M12], [M12, M22]])
```



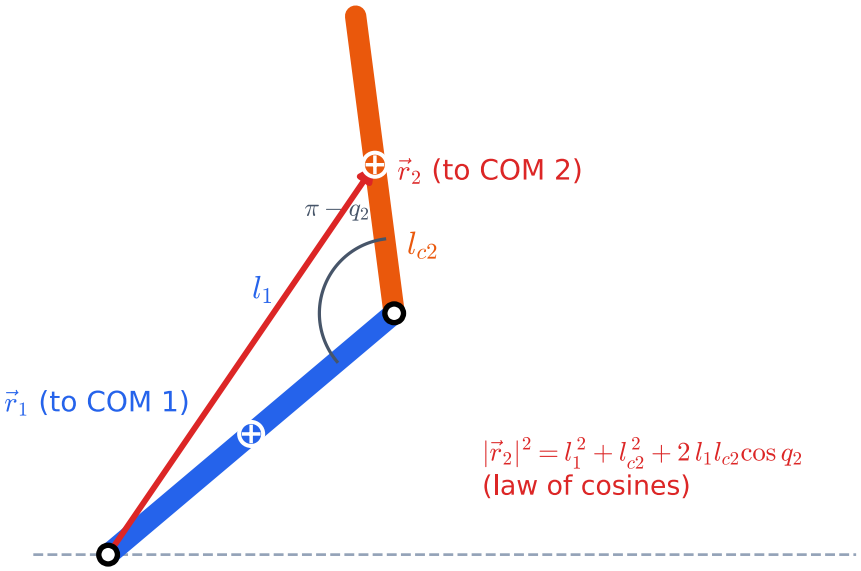

**Pose-dependent inertia.** The same arm (same masses) has a very different $M_{11}$ depending on how it is folded: mass far from joint 1 is expensive to swing. Values from the real `mass_matrix` with no payload.

We now derive these three lines from scratch. (The payload enters only through $m_{2e}, l_{c2e}, I_{2e}$ from `combine_payload`, so below I simply write those quantities; set $m_p = 0$ and they reduce to $m_2, l_{c2}, I_2$.)

## 2 Step 1: where is everything? (kinematics)

To get kinetic energy we need the **speed** of each centre of mass, so we first need its **position**. Use ordinary trigonometry: a point at distance $r$ from a pivot in direction $\alpha$ (measured from $+x$) is at $(r\cos\alpha,\ r\sin\alpha)$. Link 1 points in direction $q_1$; link 2 points in direction $q_1 + q_2$ (because $q_2$ is measured relative to link 1, the angles *add*).

$$\mathbf{r}_1 = l_{c1}\begin{pmatrix}\cos q_1\\ \sin q_1\end{pmatrix}, \qquad \mathbf{r}_2 = \underbrace{l_1\begin{pmatrix}\cos q_1\\ \sin q_1\end{pmatrix}}_{\text{elbow}} + l_{c2e}\begin{pmatrix}\cos(q_1+q_2)\\ \sin(q_1+q_2)\end{pmatrix}.$$




**Positions of the two centres of mass.** $\mathbf{r}_1$ points to COM 1; $\mathbf{r}_2$ goes to the elbow and then along link 2 to COM 2. The triangle (base, elbow, COM 2) has sides $l_1$ and $l_{c2}$ enclosing the angle $\pi - q_2$, which gives the law-of-cosines formula for $|\mathbf{r}_2|^2$ that we will meet again.

## 3 Step 2: velocities

Velocity = rate of change of position. The chain rule gives $\frac{d}{dt}\cos q_1 = -\sin q_1\,\dot q_1$ and $\frac{d}{dt}\sin q_1 = \cos q_1\,\dot q_1$, and likewise for $q_1+q_2$ with the combined rate $\dot q_1+\dot q_2$. Write $s_1=\sin q_1,\ c_1=\cos q_1,\ s_{12}=\sin(q_1+q_2),\ c_{12}=\cos(q_1+q_2)$ and $\omega_2=\dot q_1+\dot q_2$ (the absolute spin rate of link 2).

$$\dot{\mathbf r}_1 = l_{c1}\dot q_1\begin{pmatrix}-s_1\\ c_1\end{pmatrix}, \qquad \dot{\mathbf r}_2 = l_1\dot q_1\begin{pmatrix}-s_1\\ c_1\end{pmatrix} + l_{c2e}\,\omega_2\begin{pmatrix}-s_{12}\\ c_{12}\end{pmatrix}.$$

Speeds squared ($|\mathbf v|^2 = v_x^2+v_y^2$). For COM 1, since $s_1^2+c_1^2=1$: $|\dot{\mathbf r}_1|^2 = l_{c1}^2\dot q_1^2$. For COM 2, square the sum:

$$|\dot{\mathbf r}_2|^2 = l_1^2\dot q_1^2 + l_{c2e}^2\omega_2^2 + 2\,l_1 l_{c2e}\,\dot q_1\omega_2 \underbrace{\left(s_1 s_{12}+c_1 c_{12}\right)}_{=\cos\left((q_1+q_2)-q_1\right)=\cos q_2}.$$

The bracket is the cosine subtraction formula $\cos(A-B)=\cos A\cos B+\sin A\sin B$. This is the key simplification: **the absolute angle $q_1$ drops out**, only the relative angle $q_2$ survives. Physically: rotating the entire arm rigidly about the base does not change how hard it is to swing.

## 4 Step 3: kinetic energy

Each link contributes "translation of its COM" plus "spinning about its COM" (König's theorem from the *combine_payload* document):

$$T = \underbrace{\tfrac12 m_1|\dot{\mathbf r}_1|^2 + \tfrac12 I_1\dot q_1^2}_{\text{link 1}} + \underbrace{\tfrac12 m_{2e}|\dot{\mathbf r}_2|^2 + \tfrac12 I_{2e}\,\omega_2^2}_{\text{link 2 + payload}}.$$

Insert the speeds and replace $\omega_2=\dot q_1+\dot q_2$, so $\omega_2^2=\dot q_1^2+2\dot q_1\dot q_2+\dot q_2^2$ and $\dot q_1\omega_2=\dot q_1^2+\dot q_1\dot q_2$:

$$\begin{aligned} T &= \tfrac12\Big[m_1 l_{c1}^2 + I_1\Big]\dot q_1^2 \\ &+\tfrac12 m_{2e}\Big[l_1^2\dot q_1^2 + l_{c2e}^2(\dot q_1^2+2\dot q_1\dot q_2+\dot q_2^2) + 2l_1 l_{c2e}c_2(\dot q_1^2+\dot q_1\dot q_2)\Big] \\ &+\tfrac12 I_{2e}\Big[\dot q_1^2+2\dot q_1\dot q_2+\dot q_2^2\Big], \qquad c_2=\cos q_2. \end{aligned}$$

Now collect the coefficients of each of the three monomials $\dot q_1^2,\ \dot q_1\dot q_2,\ \dot q_2^2$:

| monomial | coefficient in $T$ |
|---|---|
| $\dot q_1^2$ | $\tfrac12\left[m_1 l_{c1}^2 + I_1 + m_{2e}(l_1^2+l_{c2e}^2+2l_1 l_{c2e}c_2) + I_{2e}\right]$ |
| $\dot q_1\dot q_2$ | $m_{2e}\left(l_{c2e}^2 + l_1 l_{c2e}c_2\right) + I_{2e}$ |
| $\dot q_2^2$ | $\tfrac12\left[m_{2e}l_{c2e}^2 + I_{2e}\right]$ |

## 5 Step 4: reading off the matrix

A **quadratic form** with a symmetric $2\times 2$ matrix expands as

$$\tfrac12\begin{pmatrix}\dot q_1 & \dot q_2\end{pmatrix}\begin{pmatrix}M_{11} & M_{12}\\ M_{12} & M_{22}\end{pmatrix}\begin{pmatrix}\dot q_1\\ \dot q_2\end{pmatrix} = \tfrac12 M_{11}\dot q_1^2 + M_{12}\,\dot q_1\dot q_2 + \tfrac12 M_{22}\dot q_2^2.$$

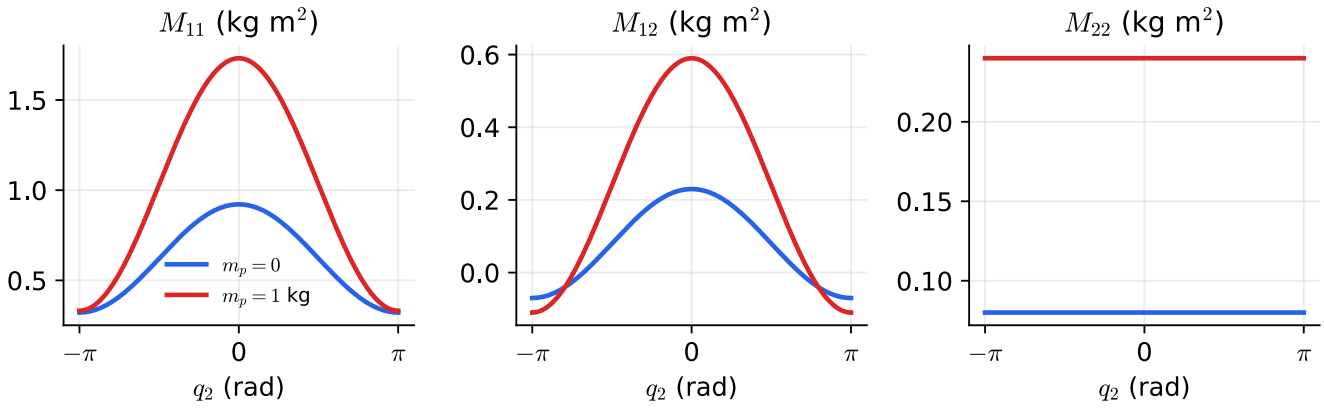
(The cross term has a 2 because $M_{12}$ appears twice, $\frac12(M_{12}+M_{21})=M_{12}$.) Compare with the table: the coefficients match if

$$\boxed{\begin{aligned} M_{11} &= m_1 l_{c1}^2 + I_1 + m_{2e}\left(l_1^2 + l_{c2e}^2 + 2 l_1 l_{c2e}\cos q_2\right) + I_{2e} \\ M_{12} &= M_{21} = m_{2e}\left(l_{c2e}^2 + l_1 l_{c2e}\cos q_2\right) + I_{2e} \\ M_{22} &= m_{2e}\, l_{c2e}^2 + I_{2e} \end{aligned}} \qquad \text{and} \qquad T = \tfrac{1}{2}\,\dot{q}^\top M(q)\,\dot{q}.$$

These are line for line the three lines in the code (`M11`, `M12`, `M22`). The matrix returned is `[[M11, M12], [M12, M22]]`, symmetric by construction.

## 6 Step 5: what the entries mean physically

- $M_{22} = m_{2e} l_{c2e}^2 + I_{2e}$: the inertia of link 2 (with payload) about the *elbow*. It is just the parallel-axis theorem: inertia about the COM plus mass times distance$^2$. It is constant because link 2 always has the same shape relative to joint 2.
- $M_{11} = \left[m_1 l_{c1}^2 + I_1\right] + \left[m_{2e}|\mathbf{r}_2|^2 + I_{2e}\right]$: the inertia of the *whole arm* about the base. First bracket: link 1 about joint 1 (parallel axis). Second bracket: link 2 about joint 1, where $|\mathbf{r}_2|^2 = l_1^2 + l_{c2e}^2 + 2 l_1 l_{c2e}\cos q_2$ is the squared distance from the base to COM 2 (law of cosines, see the kinematics figure). Stretched arm ($q_2 = 0$, $\cos = 1$): distance is largest, $M_{11}$ is largest. Folded ($q_2 = \pi$, $\cos = -1$): COM 2 is closest to the base, $M_{11}$ is smallest.
- $M_{12}$: the *coupling*. If you accelerate joint 2, link 2's reaction pushes back on joint 1. It equals $M_{22}$ plus a term $m_{2e} l_1 l_{c2e}\cos q_2$ that depends on posture.




**The entries versus $q_2$** (with $q_1 = 0.3$; $M$ does not depend on $q_1$ at all). $M_{22}$ is flat; $M_{11}$ and $M_{12}$ follow a cosine, peaking at $q_2 = 0$ (stretched) and dipping at $q_2 = \pm\pi$ (folded). Red = 1 kg payload: every entry rises.

## 7 Properties you can rely on

### Symmetric

It must be, because the kinetic energy quadratic form only ever fixes $M_{12} + M_{21}$; choosing $M_{12} = M_{21}$ is the standard convention (and the code builds it that way).

### Positive definite

Kinetic energy can never be negative, and it is zero only if nothing moves: $\frac{1}{2}\dot{q}^\top M \dot{q} > 0$ for every $\dot{q} \neq 0$. This guarantees that $M$ is **invertible**, which is essential, since the simulator must solve $M\ddot{q} = \ldots$. A direct proof: write $k = m_{2e} l_1 l_{c2e}\cos q_2$ and $D = M_{22}$, $a = m_1 l_{c1}^2 + I_1$. Then $M_{11} = a + m_{2e} l_1^2 + D + 2k$, $M_{12} = D + k$ and

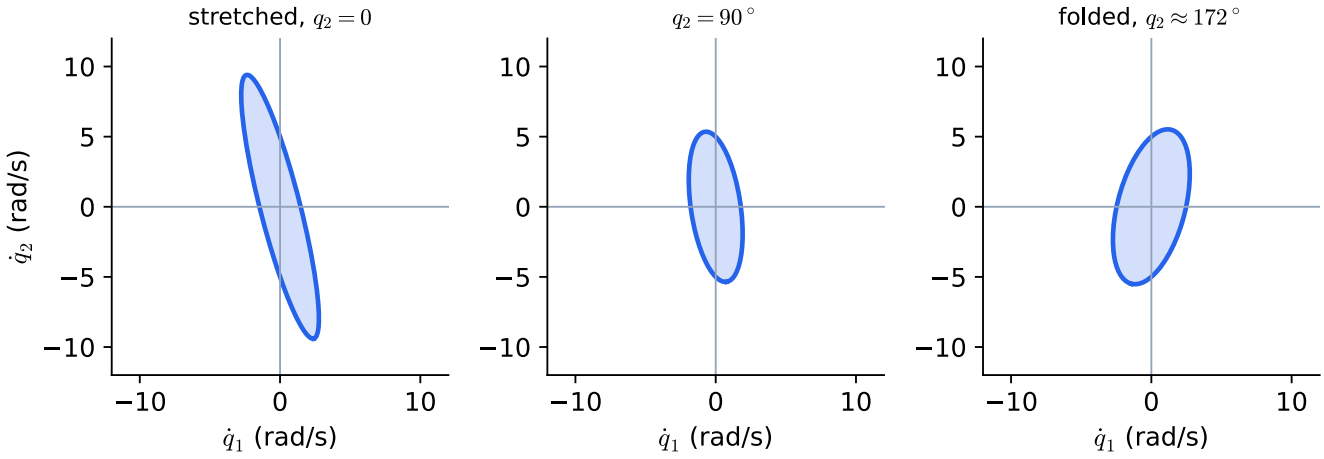
$$\det M = M_{11}M_{22} - M_{12}^2 = D\,(a + m_{2e}l_1^2) - k^2.$$

(the $D^2$ and $2kD$ terms cancel). Since $D \geq m_{2e}l_{c2e}^2$ we get $D\,m_{2e}l_1^2 \geq m_{2e}^2 l_1^2 l_{c2e}^2 \geq k^2$, and the extra $D\,a > 0$ makes the determinant strictly positive. ✓

### Independent of $q_1$

As found in Step 2: only $q_2$ appears. (Gravity is what makes the arm care about $q_1$; that is the next function.)

The positive-definiteness has a nice picture: the set of velocities with the same kinetic energy, $\frac{1}{2}\dot{q}^\top M\dot{q} = 1$ J, is an **ellipse**. Long directions = cheap motions, short directions = expensive.




**Constant-energy ellipses** $\frac{1}{2}\dot{q}^\top M\dot{q} = 1$ J for three postures (no payload). Points on an ellipse all cost the same energy, so the long axis is the "cheap" direction. The tilt is the coupling $M_{12}$. For the stretched arm the cheap direction is the two joints turning in *opposite* senses ($\dot{q}_2 \approx -\dot{q}_1$): then link 2's absolute spin $\dot{q}_1 + \dot{q}_2$ is small. Moving both joints the same way is the expensive combination.

## 8 Worked example with real numbers

Take $m_p = 0.5$ kg, $q = (0.5,\ 0.8)$ rad, so $\cos q_2 = 0.6967$. From *combine_payload*: $m_{2e} = 2.0000$, $l_{c2e} = 0.2500$, $I_{2e} = 0.03500$. Link 1 has $m_1 l_{c1}^2 + I_1 = 0.16667$.

| entry | calculation | value |
|---|---|---|
| $M_{11}$ | $0.16667 + 0.97335 + 0.03500$ | 1.17502 |
| $M_{12}$ | $0.29918 + 0.03500$ | 0.33418 |
| $M_{22}$ | $0.12500 + 0.03500$ | 0.16000 |

Check: $\det M = 0.076329 > 0$ and the eigenvalues are $0.0599, 1.2752$ (both positive), identical to what `mass_matrix(q, p)` returns.

## 9 Code ↔ maths dictionary

| code | maths |
|---|---|
| `q1, q2 = q` (q1 unused!) | $M$ depends on $q_2$ only |
| `m2e, lc2e, I2e = combine_payload(...)` | $m_{2e}, l_{c2e}, I_{2e}$ |

| | |
|---|---|
| `c2 = np.cos(q2)` | $\cos q_2$ (from $\cos(A-B)$ identity) |
| `p.m1*p.lc1**2 + p.I1` | link 1 inertia about joint 1 (parallel axis) |
| `m2e*(p.l1**2 + lc2e**2 + 2*p.l1*lc2e*c2)` | $m_{2e}\lvert\mathbf{r}_2\rvert^2$ (law of cosines) |
| `M12 appears twice in np.array` | symmetry of the quadratic form |

**CHECK YOURSELF**

1. Which entry does not change when the elbow bends? *Answer:* $M_{22}$.
2. At $q_2 = \pi$ (arm folded back on itself) with no payload and $l_1 = 0.5,\ l_{c2} = 0.2$: is $M_{12}$ larger or smaller than $M_{22}$? *Answer:* $M_{12} = m_2(l_{c2}^2 - l_1 l_{c2}) + I_2 = m_2 l_{c2}(l_{c2} - l_1) + I_2$; since $l_1 > l_{c2}$ the bracket is negative, so $M_{12} < M_{22}$ (here $1.5 \cdot 0.2 \cdot (-0.3) + 0.02 = -0.07$).
3. Why is there no $q_1$ in the function body, though $q_1$ is unpacked? *Answer:* rotating the whole arm rigidly changes no relative geometry; the cosine identity removed $q_1$.